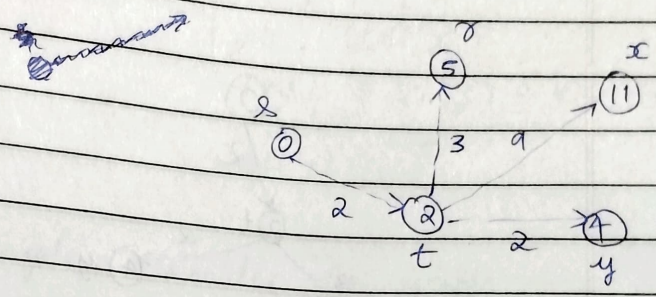


$u \leftarrow \text{extract_min}(Q) = t$
 $\text{adj}[t] \leftarrow x, s$

$\text{relax}(t, y)$	$\text{relax}(t, r)$	$\text{relax}(t, x)$
$d[y] > d[t] + w(t, y)$ $\infty > 0 + 2 = 2$ $d[y] = 2, \pi[y] = t$	$d[r] > d[t] + w(t, r)$ $10 > 0 + 3 = 3$ $d[r] = 3, \pi[r] = t$	$d[x] > d[t] + w(t, x)$ $\infty > 0 + 9 = 9$ $d[x] = 9, \pi[x] = t$

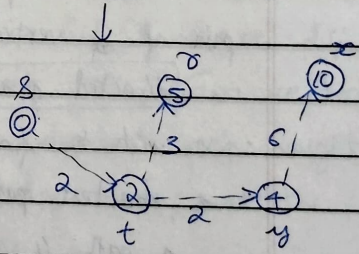


$(iv) u \leftarrow \text{extract_min}(Q) = y$
 $\text{adj}[y] \leftarrow x, s$

$\text{relax}(y, x): d[x] \# > d[y] + w(y, x) = 2 + 6 = 8$
 $\therefore d[x] = 8, \pi[x] = y$
 $\text{relax}(y, s): d[s] = 0 < d[y] + w(y, s) = 2 + 7 = 9$
 no change

$(v) u \leftarrow \text{extract_min}(Q)$
 $u = r$

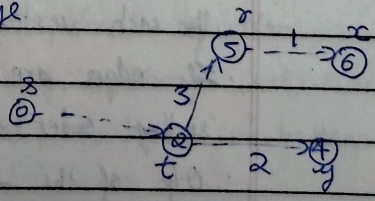
$\text{relax}(r, x): d[x] > d[r] + w(r, x)$
 $8 > 3 + 1$
 $8 > 4$
 $\therefore d[x] = 4, \pi[x] = r$

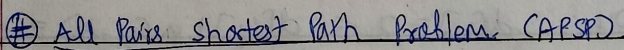
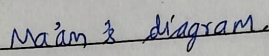


$\text{relax}(r, t): d[t] < d[r] + w(r, t) = 3 + 2 = 5$
 no change

$(vi) u = \text{extract_min}(Q) = 6$

$\text{relax}(x, y): d[y] < d[x] + w(x, y) = 4 + 4 = 8$ No change
 stop algo. All vertices considered.





• We wish to find for every pair of vertices (u, v) a shortest path from u to v where weight of a path is sum of weights of its constituent edges.

We can solve this prob. by running Single Source Shortest Path algo. V times.

If edges are +ve, Dijkstra's algo. may be applied. If some are -ve, Bellman-Ford is used, once for each vertex.

- If edges are -ve & vertices can be placed in topological order then DAG may be used.

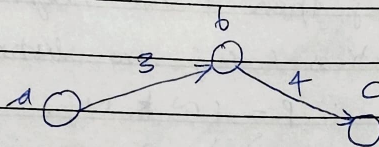
o/p of the algo. APSP prob. is represented in tabular form.

- Adjacency matrix & represent:-
 we assume that vertices of given graph can be numbered as $1, 2, 3, \dots, n$, so that, input is a $n \times n$ matrix.
- We represent the edge weights of n -vertexed directed graph G as:

$$W = w(i, j) \text{ where,}$$

$$w(i, j) = \begin{cases} 0, & i = j \\ \infty, & (i, j) \notin E \\ w_e, & \text{if } (i, j) \in E, \\ & \downarrow \\ & \text{weight of edge } (i, j) \end{cases}$$

- For example:- Consider the graph -



Adj. matrix is written as:-

	a	b	c
a	0	3	∞
b	∞	0	4
c	∞	∞	0

The alp is a $n \times n$ matrix $D = (d_{ij})$ where d_{ij} contains weight of the shortest path from vertex i to j . If δ_{ij} denotes the shortest path weight from vertex i to j then $d_{ij} = \delta_{ij}$ at the termination of the algorithm.

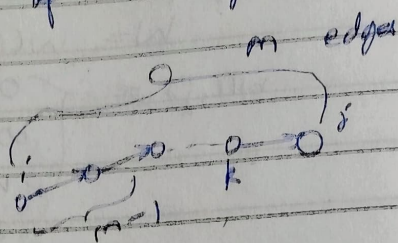
~~Ex 10~~

⊕ Matrix Multiplicⁿ method to solve APSP (dynamic programming method).

Steps used in DP/Matrix multiplicⁿ are as follows:

1. Characterize the structure of optimal solution.
2. Recursively define the value of optimal solution.
3. Compute the values of optimal solution in bottom-up fashion.

- I. Characterize the struc. of optimal solⁿ.
- We know that for APSP, it is already proved that every subpath of a shortest path are shortest.
 - Suppose that, the graph is represented by adjacency matrix $W = (w_{ij})$ and P is a ^{shortest} path from vertex i to j consisting of ' m ' edges.



- Assuming that there is no negative cycle and m is finite.
 - If $i = j$ then P has weight 0 and no edge.
 - If vertices i & j are distinct then we decompose P into : $P = i \sim k \sim j$.
- $$\Rightarrow P' \cup (k, j)$$
- ↓
 $m-1$ edges

Hence, the prop. P' is the shortest path from i to k .
 $\therefore \delta(i, j) = \delta(i, k) + w(k, j)$

II. Recursive definition.

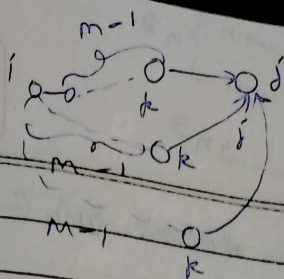
- Let $l_{ij}^{(m)}$ be the $\min^m$ weight of any path from vertex i to vertex j that contains atmost m edges, where $l_{ij} = \begin{cases} 0, & i = j. \\ \infty, & i \neq j. \end{cases}$

- For $m \geq 1$, we compute $l_{ij}^{(m)}$ as the $\min^m$ weight of $l_{ij}^{(m-1)}$ and $\min^m$ wt. obtained by looking at all possible predecessors k of j .

$$l_{ij}^{(m)} = \min \left(l_{ij}^{(m-1)}, \min_{1 \leq k \leq n} (l_{ik}^{(m-1)} + w_{kj}) \right)$$

$1 \leq k \leq n$

- Thus, we recursively define $l_{ij}^{(m)}$ as



If the graph contains no negative weight cycle then for every pair of vertices (i, j) for which $d_{ij} < \infty$ there is a shortest path from vertex i to j that is simple (vertices visited only once) and thus contains at most $n-1$ edges.

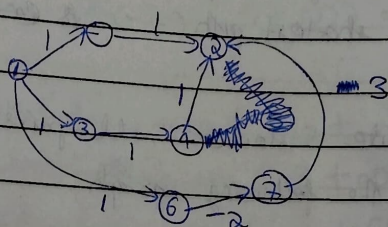
A path from vertex i to j with more than $n-1$ edges cannot have lower weight than a shortest path from vertex i to j .

$$d_{ij} = d_{ij}^{(n-1)} = d_{ij}^{(n)} = d_{ij}^{(n+1)} = \dots$$

Even

from 1 to 2

Shortest path is 2.



No other path

from 1 to 2 has wt. lesser than 2.

⊕ Computing the shortest path.

Taking as our input the matrix $W = (w_{ij})$, we now compute $L^{(1)}, L^{(2)}, \dots, L^{(n-1)}$ where $L^{(m)} = (l_{ij}^{(m)})$.

To find the matrix $L^{(m)}$, i.e. to find the shortest path weight do the following:-

extended shortest path (L, w) :-

1. $n \leftarrow$ rows of W

2. $L' \leftarrow (l'_{ij})$ // temporary matrix

3. for $i=1$ to n

4. for $j=1$ to n

5. $l'_{ij} \leftarrow \infty$

6. for $k=1$ to n

7. $l'_{ij} = \min(l'_{ij}, \min(l_{ik} + w_{kj}))$

8. return L'

$$L^{(1)} = W.$$

$$L^{(2)} = L^{(1)} \cdot W$$

$$L^{(n-1)} = L^{(n-2)} \cdot W$$

$$\begin{aligned}
 & \{i \rightarrow n\} n^2 \\
 & \{ij \rightarrow n\} n^2 \\
 & n^3
 \end{aligned}$$

Complexity of extend shortest path (...) $\rightarrow O(n^3)$

slower shortest path W :-

$$L^{(1)} \leftarrow W$$

$$n \leftarrow \text{no. of rows of } W$$

for $m=2$ to $(n-1)$:
operations

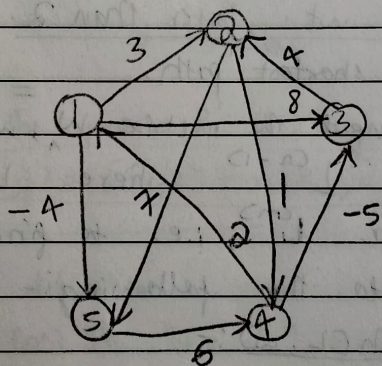
$$L^{(m)} = \text{extend-shortest-path}(L^{(m-1)}, W)$$

$$\text{return } L^{(n-1)} \quad \rightarrow O(n^3)$$

$\therefore$ Entire algo. $\rightarrow O(n^4)$

We see that slower shortest path (...) is in $O(n^4)$
is for loop is executed $(n-2)$ times & extend-
shortest path (...) is executed $O(n^3)$.

Q1. Execute slower all-pairs shortest-path algorithm on the following graph.



Step I:-

$$L^{(1)} = W =$$

	1	2	3	4	5
1	0	3	8	∞	-4
2	∞	0	∞	1	2
3	∞	4	0	∞	∞
4	2	∞	-5	0	∞
5	∞	∞	∞	6	0

Step II:- $n \leftarrow \text{no. of rows of } W = 5$

$$m=2$$

$$(1) L^{(2)} = \text{extend-shortest-path}(L^{(1)}, W)$$

$$L^j \leftarrow (l_{ij}^j)_{i,j=1}^n = \begin{matrix} & \begin{matrix} l_{11}^j & l_{12}^j & l_{13}^j & l_{14}^j & l_{15}^j \end{matrix} \\ \begin{matrix} i=1 \\ i=2 \\ i=3 \\ i=4 \end{matrix} & \begin{bmatrix} \infty & \infty & \infty & \infty & \infty \\ \infty & \infty & \infty & \infty & \infty \\ \infty & \infty & \infty & \infty & \infty \\ \infty & \infty & \infty & \infty & \infty \end{bmatrix} \end{matrix}$$

for $k=1$ to 5

$$l_{ij}^j = \min (l_{ij}^j, \min (l_{ik} + w_{kj}))$$

 $j=1$ to 5

$$l_{ij}^j = \min (l_{ij}^j, \min (l_{ik} + w_{kj}))$$

$$l_{ij}^j = l_{ij}^j, k=1$$

$$l_{11}^j = \min (l_{11}^j, \min (l_{11} + w_{11}))$$

$$= \min (0, \min (0 + 0)) = 0$$

$$(i) k=2$$

$$l_{12}^j = \min (l_{12}^j, \min (l_{12} + w_{21}))$$

$$= \min (0, \min (0 + 3)) = 0$$

$$(ii) j=1, k=3, l_{11}^j = \min (l_{11}^j, \min (l_{13} + w_{31})) = 0$$

$$0, \min (8 + \infty)$$

$$(v) j=1, k=4, l_{11}^j = \min (l_{11}^j, \min (l_{14} + w_{41})) = 0$$

$$(v) j=1, k=5, l_{11}^j = \min (l_{11}^j, \min (l_{15} + w_{51})) = 0$$

$$0, \min (-4 + \infty)$$

$$l_{11}^j = \min (l_{11}^j, \min (l_{11} + w_{11}), (l_{12} + w_{21}), (l_{13} + w_{31}), (l_{14} + w_{41}), (l_{15} + w_{51}))$$

$$\therefore l_{11}^j = 0$$

$$2. j=2$$

$$k=1$$

$$k=2$$

$$k=3$$

$$l_{12}^j = \min (l_{12}^j, \min (l_{11} + w_{12}, l_{12} + w_{22}, l_{13} + w_{32}, l_{14} + w_{42}, l_{15} + w_{52}))$$

$$= \min (0, \min (0 + 3, 3 + 0, 8 + 4, \infty + \infty, -4 + \infty))$$

$$= \min (0, \min (3, 3, 12, \infty, \infty)) = 3 \therefore l_{12}^j = 3$$

$$3. j=3$$

$$k=1$$

$$k=2$$

$$k=3$$

$$k=4$$

$$l_{13}^j = \min (l_{13}^j, \min (l_{11} + w_{13}, l_{12} + w_{23}, l_{13} + w_{33}, l_{14} + w_{43}, l_{15} + w_{53}))$$

$$= \min (0, \min (0 + 8, 3 + \infty, 8 + 0, \infty + \infty, -4 + \infty))$$

$$= \min (0, \min (8, \infty, 8, \infty, \infty)) = 8$$

$L^2 \Rightarrow$ at most 2 edges allowed b/w any 2 vertices.

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$$j=4, \quad k=1 \quad k=2 \quad k=3$$

$$l'_{14} = \min(l'_{14}, \min_{k=4} (l_{11} + w_{14}, l_{12} + w_{24}, l_{13} + w_{34}, l_{14} + w_{44}, l_{15} + w_{54}))$$

$$= \min(\infty, \min(\infty + \infty, 3+1, 8+\infty, \infty, -4+6))$$

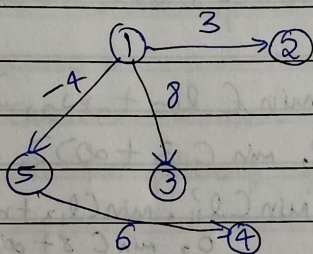
$$= \min(\infty, \min(\infty, 4, \infty, \infty, 2)) = 2.$$

$$j=5, \quad k=1 \quad k=2 \quad k=3$$

$$l'_{15} = \min(l'_{15}, \min_{k=4} (l_{11} + w_{15}, l_{12} + w_{25}, l_{13} + w_{35}, l_{14} + w_{45}, l_{15} + w_{55}))$$

$$= \min(\infty, \min(\infty + (-4), 3+7, 8+\infty, \infty, -4+\infty))$$

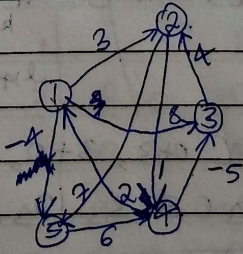
$$= \min(\infty, \min(-4, 10, \infty, \infty, -4)) = -4.$$



$$L^2 = \begin{bmatrix} 0 & 3 & 8 & 2 & -4 \\ \infty & \infty & \infty & \infty & \infty \\ \infty & \infty & \infty & \infty & \infty \\ \infty & \infty & \infty & \infty & \infty \\ \infty & \infty & \infty & \infty & \infty \end{bmatrix}$$

Similarly find $l'_{2j}, l'_{3j}, l'_{4j}, l'_{5j}$ to get $L^{(2)}$

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$$L^{(3)} = \begin{bmatrix} 0 & 3 & 8 & 2 & -4 \\ 3 & 0 & -4 & 1 & -1 \\ 8 & -4 & 0 & 5 & 11 \\ 2 & -1 & -5 & 0 & -2 \\ 8 & 5 & 1 & 6 & 0 \end{bmatrix}$$


$$L^{(2)} = \begin{bmatrix} 0 & 3 & 8 & 2 & -4 \\ 3 & 0 & -4 & 1 & -1 \\ \infty & \infty & 0 & 5 & 11 \\ 2 & -1 & -5 & 0 & -2 \\ 8 & 5 & 1 & 6 & 0 \end{bmatrix}$$

$$L^{(4)} =$$

1	0	1	-3	7	5
2	2	0	-4	2	-4
3	7	4	0	5	3
4	2	-1	-5	0	-2
5	8	5	1	6	0

Write $L^{(4)}$
 directly. For few terms of 1st row show using formula

Ex: $L_{12}^{(4)} = \min(C_3, 8+4, \dots)$
 edge² edges³ edges⁴

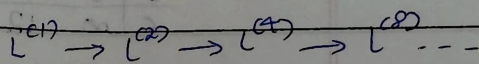
Repeated Squaring

$$L^{(n)} = L^{(2^{\log_2 n})}$$

$$L^{(n-1)} = L^{(2^{\log_2(n-1)})} \Rightarrow L^{(2^{\log_2(n-1)})}$$

$$\begin{bmatrix} L^{(1)} \\ L^{(2)} = L^{(1)} L^{(1)} \\ L^{(4)} = L^{(2)} L^{(2)} \\ L^{(8)} = L^{(4)} L^{(4)} \\ \vdots \end{bmatrix}$$

Repeated Squaring method:-



- We know we use repeated squaring method to reduce the complexity which now uses seq. of matrices by using repeated squaring method.
- Each of $\log_2(n-1)$ matrix takes n^3 time.
- Total time $\rightarrow O(n^3 \log_2 n)$

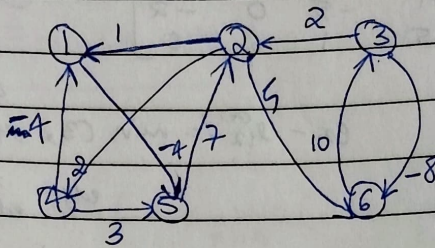
Faster all-pairs (C_n) :-

1. $n \leftarrow \text{rows}(C_n)$
2. $L^{(1)} \leftarrow C_n$
3. $m \leftarrow 1$
4. while $(m < (n-1))$ do
5. $L^{(2^m)} \leftarrow \text{extend_shortest_path}(L^{(2^{m-1})}, L^{(2^{m-1})})$
 $m \leftarrow 2m$
6. return $L^{(2^m)}$

while loop executes $\log_2(n-1)$ times

Note:- In order to compute $L^{(n-1)}$ we compute $L^{(2)}, L^{(4)}, L^{(8)}, \dots, L^{(n-1)}$

Q. Execute faster all pairs shortest path algo. on the following graph from every vertex to every other vertex.



$$L^{(0)} = \begin{array}{c|cccccc} & 1 & 2 & 3 & 4 & 5 & 6 \\ \hline 1 & 0 & 3 & \infty & \infty & -4 & \infty \\ 2 & -2 & 0 & 15 & 2 & -3 & 5 \\ 3 & 3 & 2 & 0 & 4 & \infty & -8 \\ 4 & -4 & 10 & \infty & 0 & -8 & \infty \\ 5 & 8 & 7 & 2 & 9 & 0 & 12 \\ 6 & \infty & 12 & 10 & \infty & \infty & 0 \end{array}$$

$$L^{(1)} = \begin{array}{c|cccccc} & 1 & 2 & 3 & 4 & 5 & 6 \\ \hline 1 & & & & & & \\ 2 & & & & & & \\ 3 & & & & & & \\ 4 & & & & & & \\ 5 & & & & & & \\ 6 & & & & & & \end{array}$$